Aug 7 2025
BDBP106: Linux and Python programming Lab 6

Learning goals: Linux OS commands with piping, shell types, parent/child relationships, internal and external commands, environment variables, PATH variable, variable arrays and other commands not seen in theory classes
NOTE: Save screenshots of each exercise, and upload your work to your github account as Lab6.pdf by end of Thursday Aug 7.
EXERCISES Start a fresh terminal for this Lab.

(1) The echo command can be used simply to display a piece of text (useful in programs). Use echo to create a sentence. Pipe the output of this command to count the number of words in the sentence. How do I use the wc to count the number of lines in this sentence?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ echo Sumirtha Joyce is best|wc -l
1
ibab@IBAB-MSC-BDB-comp16-ibab:~$ echo Sumirtha Joyce is best|wc -w
4
```

Use wc -l to count the number of lines in sentence
Use wc -w to count the words in the sentence

(2) Execute the following command, and explain the meaning of the output. This concerns the appearance of the command prompt, and you should be able to dissect the output completely and explain it part by part. Make a list of the parts of the output and write your explanations against each part. The command is cat .bashrc | grep ’PS1’

cat .bashrc: Displays the contents of the shell configurtation file
|: Pipes the output to the next command
grep ‘PS1’: Filters lines containing ‘PS1’
${debian_chroot:+($debian_chroot)} : a parameter expansion feature in bash that serves a specific purpose in debian based sysytem
\[.. \] :marker for the color codes

\033[01;32m\] :terminal control code to set the output color where 32m outputs green color

\u@\h : username, hostname

\w : current working directory

\$:a dollar sign, unless root, in which case a hash sign

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cat .bashrc | grep 'PS1'
    PS1='${debian_chroot:+($debian_chroot)}\[\033[01;32m\]\u@\h\[\033[00m\]:\[\033[01;34m\]\w\[\033[00m\]\$ '
    PS1='${debian_chroot:+($debian_chroot)}\u@\h:\w\$ '
    PS1="\[\e]0;${debian_chroot:+($debian_chroot)}\u@\h: \w\a\]$PS1"
```

(3) In the above exercise, filter the output based on ’HIST’ pattern. You will see a list of environment variables. Figure out what they stand for and what the current values mean.

Aug 7 2025
BDBP106: Linux and Python programming Lab 6

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cat .bashrc | grep 'HIST'
HISTCONTROL=ignoreboth
# for setting history length see HISTSIZE and HISTFILESIZE in bash(1)
HISTSIZE=1000
HISTFILESIZE=2000
```

HISTSIZE :Maximum number of commands stored in memory ; 1000 recent commands are available in the current session
HISTFILESIZE: Maximum size of history file ; 2000 are saved to disk for preference
HISTCONTROL: Controls which commands are saved ; eliminates duplicate entries for cluttering
(4) The whereis command searches for a program in a predefined set of standard binary directories such as /bin, /usr/bin and /usr/sbin. Type whereis ls and study the output. What is the difference between which and whereis commands?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ whereis ls
ls: /usr/bin/ls /usr/share/man/man1/ls.1.gz
ibab@IBAB-MSC-BDB-comp16-ibab:~$ which ls
/usr/bin/ls
```

Whereis: locates all resourses ensuring system wide program resource discovery: searches predefined system locations regardless of PATH
which: finds the exact location of an executable; checking if a command exists in PATH and only looks in directories listed in PATH

(5) The command dirname is used to retrieve the directory name in a given file path. Navigate to a previous lab folder such as Lab4 and list the files. Then type the command dirname <filename> where give some existing filename in this command. What is the output?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd Downloads
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ dirname lab5/lowkival
lab5
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$
```

(6) Create a local variable called mylabdir, and set it to the following value: /home/ibab/Lab6 (which means that if you have not created a Lab6 folder you should do so). Print the value of this variable using the echo command, and then the dirname command output with a file you are working with, under the Lab6 directory.

Aug 7 2025
BDBP106: Linux and Python programming Lab 6

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ mylabdir=/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$ echo $mylabdir
/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$ dirname lab6
.
ibab@IBAB-MSC-BDB-comp16-ibab:~$ cd lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab6$ touch prac
ibab@IBAB-MSC-BDB-comp16-ibab:~/lab6$ cd
ibab@IBAB-MSC-BDB-comp16-ibab:~$ dirname mylabdir/prac
mylabdir
ibab@IBAB-MSC-BDB-comp16-ibab:~$ dirname $mylabdir/prac
/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(7) Create a new bash subshell in this terminal. Prove that you have created a subshell using the ps command with the appropriate options. Inside this subshell print the value of the variable mylabdir. What is the value? If it is empty, how do you convert it to a global variable? Do the conversion and show that mylabdir is indeed a global variable.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ bash
ibab@IBAB-MSC-BDB-comp16-ibab:~$ ps
    PID TTY          TIME CMD
   7535 pts/1    00:00:00 bash
   8045 pts/1    00:00:00 bash
   8052 pts/1    00:00:00 ps
ibab@IBAB-MSC-BDB-comp16-ibab:~$ ps --forest
    PID TTY          TIME CMD
   7535 pts/1    00:00:00 bash
   8045 pts/1    00:00:00  \_ bash
   8055 pts/1    00:00:00      \_ ps
ibab@IBAB-MSC-BDB-comp16-ibab:~$ echo $mylabdir

ibab@IBAB-MSC-BDB-comp16-ibab:~$ export mylabdir=/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$ echo $mylabdir
/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$ env |grep mylabdir
mylabdir=/home/ibab/lab6
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(8) Display the output of the command echo $PATH. Is PATH local or global variable? Describe two ways of finding this out.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
ibab@IBAB-MSC-BDB-comp16-ibab:~$ env | grep PATH
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
ibab@IBAB-MSC-BDB-comp16-ibab:~$
```

(9) The concept of a directory stack. Learning about this will allow you to know the powerful navigation mechanisms in Linux. The key commands in this exercise are dirs,cd -, pushd and popd. The directory stack operates like a stack of plates – the last plate on

Aug 7 2025
BDBP106: Linux and Python programming Lab 6

top is the first to be taken out. Try the following exercises to understand this.
(i) Execute dirs -v -l. What is the output? Learn more about the command and the options by looking up the man pages of what these options did. For each command below execute dirs -v -l to understand what happened to the directory stack.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ dir -v -l
total 60
drwxr-xr-x 3 ibab ibab 4096 Aug  5 11:09 Desktop
drwxr-xr-x 2 ibab ibab 4096 Jul 16 16:21 Documents
drwxr-xr-x 4 ibab ibab 4096 Aug  6 23:27 Downloads
drwxrwxr-x 7 ibab ibab 4096 Aug  1 14:48 JavaMorph
drwxrwxr-x 3 ibab ibab 4096 Jul 17 15:06 MEGA\ X
drwxr-xr-x 2 ibab ibab 4096 Jul 16 16:21 Music
drwxrwxr-x 3 ibab ibab 4096 Aug  6 09:13 Pictures
drwxr-xr-x 2 ibab ibab 4096 Jul 16 16:21 Public
drwxr-xr-x 2 ibab ibab 4096 Jul 16 16:21 Templates
drwxr-xr-x 2 ibab ibab 4096 Jul 16 16:21 Videos
drwxrwxr-x 2 ibab ibab 4096 Aug  7 14:14 bin
drwxrwxr-x 3 ibab ibab 4096 Jul 17 14:59 igv
drwxrwxr-x 2 ibab ibab 4096 Aug  7 16:06 lab6
drwx------ 4 ibab ibab 4096 Jul 17 11:12 snap
drwxrwxr-x 2 ibab ibab 4096 Aug  3 19:02 sums
```

(ii) By default the directory stack contains only one entry – the path to your $HOME.
Let's add some entries to this stack using the following commands:
(a) pushd /var/log
(b) pushd /tmp
(c) pushd /etc
(d) pushd ~/Downloads
(e) pushd ~/Documents

```
ibab@IBAB-MSC-BDB-comp16-ibab:~$ dirs -v -l
 0  /home/ibab
ibab@IBAB-MSC-BDB-comp16-ibab:~$ pushd/var/log
bash: pushd/var/log: No such file or directory
ibab@IBAB-MSC-BDB-comp16-ibab:~$ pushd /var/log
/var/log ~
ibab@IBAB-MSC-BDB-comp16-ibab:/var/log$ pushd /tmp
/tmp /var/log ~
ibab@IBAB-MSC-BDB-comp16-ibab:/tmp$ pushd /etc
/etc /tmp /var/log ~
ibab@IBAB-MSC-BDB-comp16-ibab:/etc$ pushd ~/Downloads
~/Downloads /etc /tmp /var/log ~
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ pushd ~/Documents
~/Documents ~/Downloads /etc /tmp /var/log ~
```

(iii) Now execute dirs -v -l again. Explain the output.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Documents$ dirs -v -l
 0  /home/ibab/Documents
 1  /home/ibab/Downloads
 2  /etc
 3  /tmp
 4  /var/log
 5  /home/ibab
ibab@IBAB-MSC-BDB-comp16-ibab:~/Documents$
```

Aug 7 2025
BDBP106: Linux and Python programming Lab 6

(iv) Execute the command pushd +1. Explain what happened.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Documents$ pushd +1
~/Downloads /etc /tmp /var/log ~ ~/Documents
```

(v) Execute cd -. What happened?

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Downloads$ cd -
/home/ibab/Documents
```

(vi) Execute cd /tmp. What happened? In this case, note that we simply changed directory to an entry in the directory stack without reference to the stack in any way.

```
ibab@IBAB-MSC-BDB-comp16-ibab:~/Documents$ cd /tmp
ibab@IBAB-MSC-BDB-comp16-ibab:/tmp$
```

(vii) Execute popd. What happened? popd is responsible for removing the topmost "plate" in the stack and changes to the directory entry against index 1.

```
ibab@IBAB-MSC-BDB-comp16-ibab:/tmp$ popd
/etc /tmp /var/log ~ ~/Documents
```

(viii) Execute popd +2. What happened? How is this different from pushd +2?

```
ibab@IBAB-MSC-BDB-comp16-ibab:/etc$ pushd +2
~ ~/Documents /etc /tmp
```

```
ibab@IBAB-MSC-BDB-comp16-ibab:/etc$ popd +2
/etc /tmp ~ ~/Documents
```

(ix) In all the above, what was happening the entry that was indexed against '0'? Explain what this entry represents.